_eda_price_2025_Q1

April 7, 2026

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import subprocess
import os
from scipy.stats import chi2_contingency
from datetime import timedelta

from etl.customer import aggregate_customer
from etl.subsidy import report_customer_conversion_by_subsidy_issue_status,␣
↪report_customer_metrics_by_subsidy_issue_status
from etl.subsidy import visualize_conversion_by_subsidy_issues,␣
↪report_customer_subsidy_issues_by_product
from etl.subsidy import visualize_customer_subsidy_issues_by_product,␣
↪visualize_policy_suspension_impact
from etl.price import visualize_conversion_by_price_sweet_spots,␣
↪visualize_conversion_by_price_sweet_spots_variant
from etl.price import report_product_conversion_by_price_quartile,␣
↪visualize_conversion_by_equipment_category_price
from etl.price import visualise_conversion_by_brand_price,␣
↪show_two_panel_conversion_by_brand_price
from etl.price import visualize_conversion_by_equipment_type_brand_price,␣
↪show_conversion_by_price_dashboard,␣
↪show_product_type_price_conversion_heatmap
from etl.season import visualize_conversion_by_season,␣
↪visualize_regional_seasonality
from etl.period import visualize_conversion_by_year,␣
↪report_best_and_worst_months, show_conversion_by_price_over_time,␣
↪overlay_all_years_on_one_plot
from etl.period import visualize_conversion_by_prices_sweet_spot_shift,␣
↪visualize_decision_times, show_decision_time_dashboard
from etl.product import visualise_heat_pump_performance,␣
↪show_product_type_by_brand_heatmap, show_product_type_by_agency_heatmap,␣
↪show_brand_by_product_type_heatmap
```

```python
import warnings
warnings.filterwarnings('ignore')

df_clean = pd.read_csv('cleaned_quote_data.csv')
df_clean['dt_creation_devis'] = pd.to_datetime(df_clean['dt_creation_devis'])
print(f"Loaded data: {len(df_clean):,} quotes from {df_clean['numero_compte'].
 ↪nunique():,} customers")
```

```
Loaded data: 37,008 quotes from 25,716 customers
```

```python
customers = aggregate_customer(df_clean)
```

```
Quote-level columns created:
  - out_of_pocket: 34,991 values
  - subsidy_issue: 9,076 issues (24.5%)
  - brand-product pairs: 146 unique combinations

================================================================================
BRAND-PRODUCT VALIDATION
================================================================================

  ATLANTIC + Stove customers (should be 0): 0
  FIXED: No ATLANTIC + Stove combinations found!

  Customer dataset created: 25,716 customers
  Total columns: 61
  Key columns now available:
    - main_brand: from brand-product pair (most common option)
    - main_equipment_category: from brand-product pair (most common option)
    - brand_product_nunique: 1.2 avg options per customer
    - shopped_around: 4,484 customers (17.4%)
    - decision_days: from customer_duration_days
    - price_cv: from std_out_of_pocket / avg_out_of_pocket
    - quote_count: alias for total_quotes
    - price_range: 1142 avg
    - price_volatility: 0.26 avg

Subsidy issue type distribution:
subsidy_issue_type
No Issues            18660
Both                  2480
CEE Only              2361
MaPrimeRénov Only     2215
Name: count, dtype: int64

  Saved to 'customer_master_data.csv'
```

```python
all_customers = pd.read_csv('customer_master_data.csv')
```

```python
# Using quarter column (if you have it)
customers_2025_q1 = all_customers[
    (all_customers['year'] == 2025) &
    (all_customers['quarter'] == 1)
]

# Using date range
customers_2025_q1 = all_customers[
    (all_customers['first_quote_date'] >= '2025-01-01') &
    (all_customers['first_quote_date'] <= '2025-03-31')
]

print(f"Q1 2025 customers: {len(customers_2025_q1):,}")
```

```
Q1 2025 customers: 2,503
```

```python
customers = customers_2025_q1
```

```python
conv_no_issue, conv_with_issue =␣
 ↪report_customer_conversion_by_subsidy_issue_status(customers)
report_customer_metrics_by_subsidy_issue_status(customers)
visualize_conversion_by_subsidy_issues(customers, conv_no_issue,␣
 ↪conv_with_issue)
visualize_policy_suspension_impact(customers, price_var='max_out_of_pocket')

top_products, no_issue_pct, issue_pct =␣
 ↪report_customer_subsidy_issues_by_product(customers)
visualize_customer_subsidy_issues_by_product(customers, top_products,␣
 ↪no_issue_pct, issue_pct)

visualize_conversion_by_price_sweet_spots(customers,␣
 ↪price_var='max_out_of_pocket')
visualize_conversion_by_price_sweet_spots_variant(customers,␣
 ↪price_var='max_out_of_pocket')
report_product_conversion_by_price_quartile(customers)
visualize_conversion_by_equipment_category_price(customers,␣
 ↪price_var='max_out_of_pocket')
visualise_conversion_by_brand_price(customers, price_var='max_out_of_pocket')
show_two_panel_conversion_by_brand_price(customers,␣
 ↪price_var='max_out_of_pocket')
visualize_conversion_by_equipment_type_brand_price(customers,␣
 ↪price_var='max_out_of_pocket')

visualize_conversion_by_season(customers, price_var='max_out_of_pocket')
visualize_regional_seasonality(customers, price_var='max_out_of_pocket')
```

```
visualize_conversion_by_year(customers, price_var='max_out_of_pocket')
report_best_and_worst_months(customers, price_var='max_out_of_pocket')
show_conversion_by_price_over_time(customers, price_var='max_out_of_pocket')
overlay_all_years_on_one_plot(customers, price_var='max_out_of_pocket')
visualize_conversion_by_prices_sweet_spot_shift(customers,␣
 ↪price_var='max_out_of_pocket')
visualize_decision_times(customers, price_var='max_out_of_pocket')
show_product_type_price_conversion_heatmap(customers,␣
 ↪price_var='max_out_of_pocket')
show_decision_time_dashboard(customers, price_var='max_out_of_pocket')

visualise_heat_pump_performance(customers, price_var='max_out_of_pocket')

show_conversion_by_price_dashboard(customers, price_var='max_out_of_pocket')

show_product_type_by_brand_heatmap(customers)
show_product_type_by_agency_heatmap(customers)
show_brand_by_product_type_heatmap(customers)
```

```
Customer conversion by subsidy issue status:

                    conversion_rate  customer_count  converted_count
had_subsidy_issue
False                      0.378575            1923            728.0
True                       0.455172             580            264.0

Chi-square p-value: 1.124999e-03
Statistically significant: YES

Conversion without issues: 37.86%
Conversion with issues: 45.52%
Difference: 7.7 percentage points
Customer-level finding: +7.7 points

================================================================================
Customer characteristics by subsidy issue status
================================================================================

Metric comparison:

total_quotes:
  No issues: 1.39
  With issues: 1.53
  Ratio: 1.11x

avg_quote_amount:
```

```
  No issues: 6147.34
  With issues: 9288.33
  Ratio: 1.51x

max_quote_amount:
  No issues: 6546.66
  With issues: 10213.52
  Ratio: 1.56x

customer_duration_days:
  No issues: 14.63
  With issues: 19.80
  Ratio: 1.35x

price_range:
  No issues: 825.92
  With issues: 1730.74
  Ratio: 2.10x

multiple_quotes:
  No issues: 0.28
  With issues: 0.37
  Ratio: 1.31x

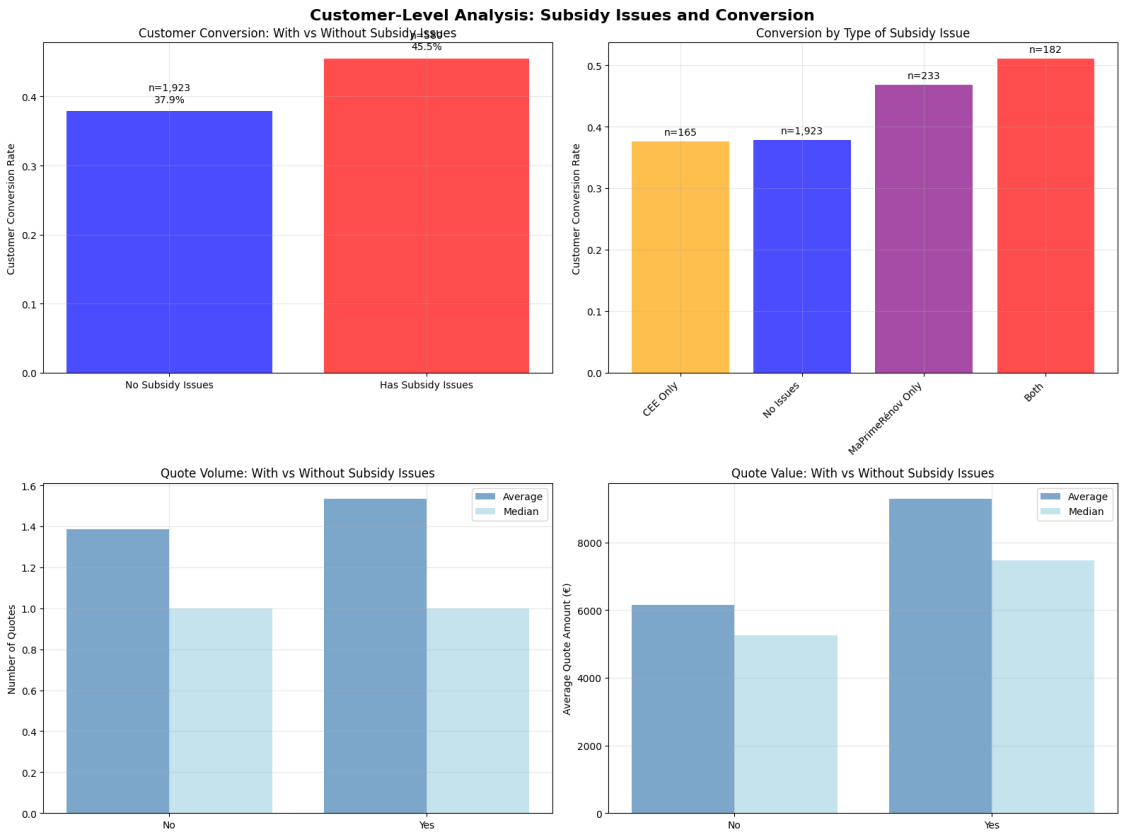
Quote pattern distribution by subsidy issue:
had_subsidy_issue  False  True
quote_pattern
Single quote       0.721  0.634
Three+ quotes      0.066  0.102
Two quotes         0.214  0.264

Decision speed by subsidy issue:
had_subsidy_issue   False  True
decision_speed
Long (>30 days)     0.074  0.103
Medium (8-30 days)  0.032  0.038
Quick (2-7 days)    0.050  0.081
Same day decision   0.844  0.778

Product family by subsidy issue status:
had_subsidy_issue                 False  True
main_product_family
Accessoire de pose                0.007  0.002
Appareil hybride                  0.003  0.022
Autres                            0.011  0.003
Chaudière                         0.425  0.071
Climatisation                     0.158  0.036
ECS : Chauffe-eau ou adoucisseur  0.072  0.033
```

```
Emetteur de chauffage  ou chappe  0.018  0.009
Photovoltaïque                    0.035  0.010
Plomberie Sanitaire               0.002  0.000
Pompe à chaleur                   0.071  0.347
Poêle                             0.177  0.467
Produit VMC                       0.005  0.000
Unknown                           0.017  0.000

Conversion by subsidy issue type:
                        mean  count
subsidy_issue_type
Both                 0.510989    182
MaPrimeRénov Only    0.467811    233
No Issues            0.378575   1923
CEE Only             0.375758    165
```

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================================================================================
Policy Suspension Impact
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Analyzing 2,414 customers
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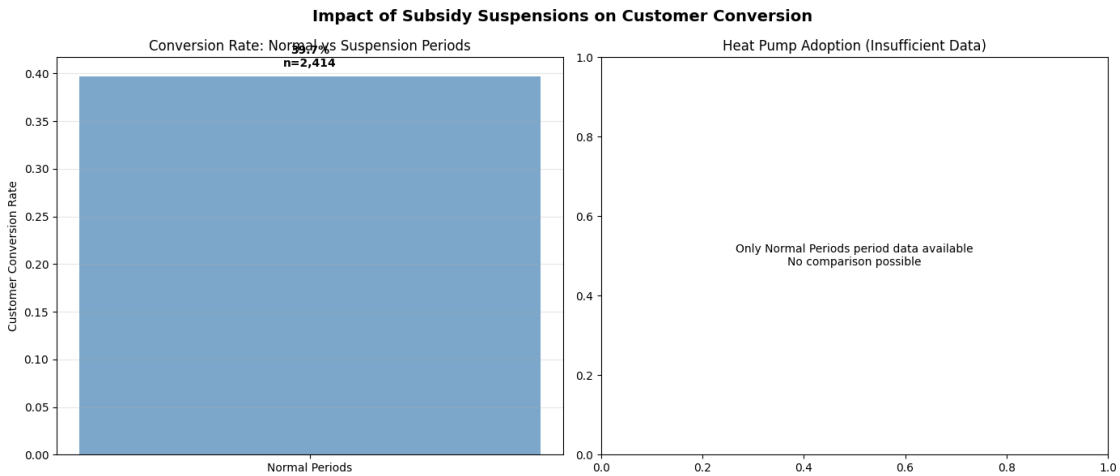
```
Price range: €1,312 to €26,385
Median price: €6,237
Unique values in during_suspension: [False]

   Only data during normal periods available

Conversion during subsidy suspensions:
                    mean  count
Normal Periods  0.397266   2414
```



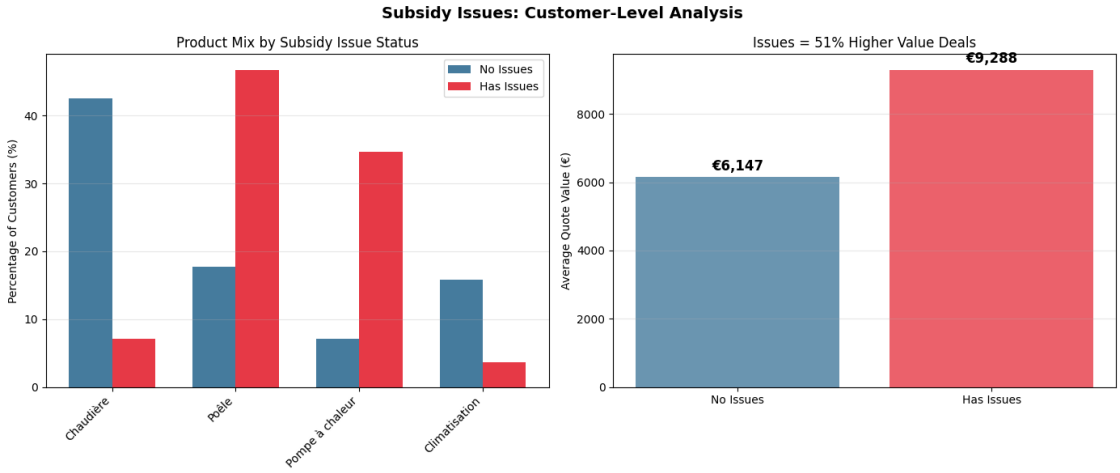

```
Unique values in main_product_family:
main_product_family
Chaudière                           858
Poêle                               612
Pompe à chaleur                     337
Climatisation                       325
ECS : Chauffe-eau ou adoucisseur    157
Photovoltaïque                       73
Emetteur de chauffage  ou chappe     39
Unknown                              33
Autres                               24
Appareil hybride                     19
Accessoire de pose                   14
Produit VMC                           9
Plomberie Sanitaire                   3
Name: count, dtype: int64

Top products:
main_product_family
Chaudière                           858
Poêle                               612
```

```
Pompe à chaleur                    337
Climatisation                      325
ECS : Chauffe-eau ou adoucisseur   157
Photovoltaïque                      73
Emetteur de chauffage  ou chappe    39
Unknown                             33
Autres                              24
Appareil hybride                    19
Name: count, dtype: int64

Using these products: ['Chaudière', 'Poêle', 'Pompe à chaleur', 'Climatisation']

No Issue %: [np.float64(42.48569942797712), np.float64(17.732709308372335),
np.float64(7.072282891315653), np.float64(15.80863234529381)]
Has Issue %: [np.float64(7.06896551724138), np.float64(46.724137931034484),
np.float64(34.6551724137931), np.float64(3.620689655172414)]
```

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================================================================================
Threshold effect
================================================================================

Analyzing 2,414 customers
Price range: €1,312 to €26,385
Median price: €6,237
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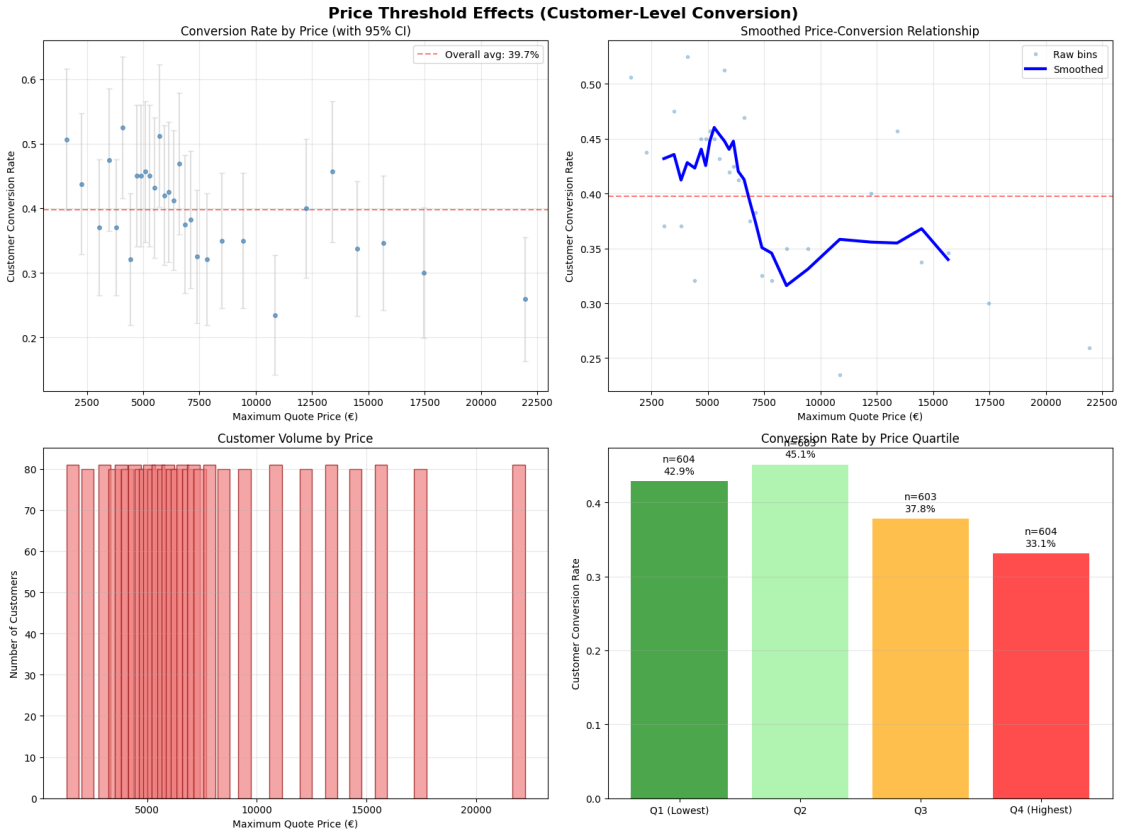


```
================================================================================
THRESHOLD EFFECT: KEY FINDINGS
================================================================================

Conversion by Price Quartile:
  Q1 (Lowest): 42.9% (n=604)
  Q2: 45.1% (n=603)
  Q3: 37.8% (n=603)
  Q4 (Highest): 33.1% (n=604)

Difference between lowest and highest quartile: 9.8 percentage points

Sweet spots (above-average conversion):
  €1,591: 50.6% (n=81)
  €2,277: 43.8% (n=80)
  €3,500: 47.5% (n=80)
  €4,084: 52.5% (n=80)
  €4,707: 45.0% (n=80)
  €4,903: 45.0% (n=80)
  €5,088: 45.7% (n=81)
```

```
  €5,282: 45.0% (n=80)
  €5,498: 43.2% (n=81)
  €5,738: 51.2% (n=80)
  €5,943: 42.0% (n=81)
  €6,132: 42.5% (n=80)
  €6,350: 41.2% (n=80)
  €6,608: 46.9% (n=81)
  €12,243: 40.0% (n=80)
  €13,398: 45.7% (n=81)

Analyzing 2,414 customers
Price range: €1,312 to €26,385
Median price: €6,237

================================================================================
Product mix by price quartile
================================================================================

Product distribution by price quartile (%):
main_equipment_category    AC  Boiler  Heat Pump  Other  Stove
price_quartile
Q1_Low                   11.5    43.2        0.8   34.7    9.7
Q2_MidLow                 9.4    40.7        1.3    4.1   44.5
Q3_MidHigh               12.0    41.0        1.6    5.5   39.9
Q4_High                  17.6    11.2       51.5   15.3    4.4

================================================================================
Conversion rates by product and price quartile
================================================================================

Heat Pump (n=359):
  Q1_Low: 40.0%
  Q2_MidLow: 50.0%
  Q3_MidHigh: 60.0%
  Q4_High: 38.2%

Boiler (n=840):
  Q1_Low: 42.9%
  Q2_MidLow: 44.6%
  Q3_MidHigh: 42.3%
  Q4_High: 29.0%

AC (n=314):
  Q1_Low: 40.8%
  Q2_MidLow: 22.4%
  Q3_MidHigh: 36.5%
  Q4_High: 32.4%
```

```
Stove (n=622):
  Q1_Low: 28.3%
  Q2_MidLow: 47.8%
  Q3_MidHigh: 33.7%
  Q4_High: 22.2%

================================================================================
HEAT PUMP CUSTOMERS (n=419)
================================================================================
Average price: €15,571
Conversion rate: 37.9%
Subsidy issue rate: 61.3%

Heat pump adoption by quartile:
  Q1_Low: 1.3%
  Q2_MidLow: 1.3%
  Q3_MidHigh: 2.1%
  Q4_High: 60.3%

================================================================================
Equipment Category Segment Effect
================================================================================

Analyzing 2,414 customers
Price range: €1,312 to €26,385
Median price: €6,237
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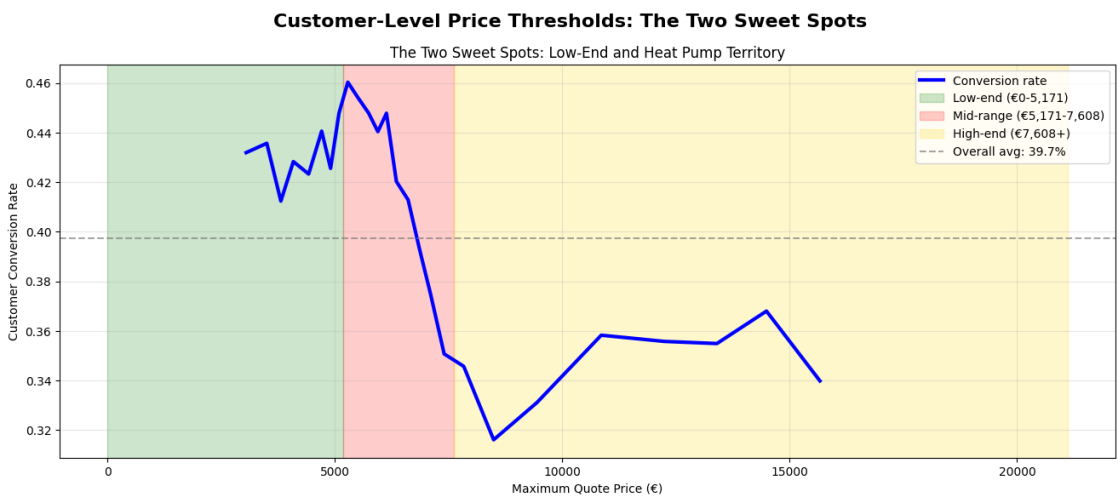

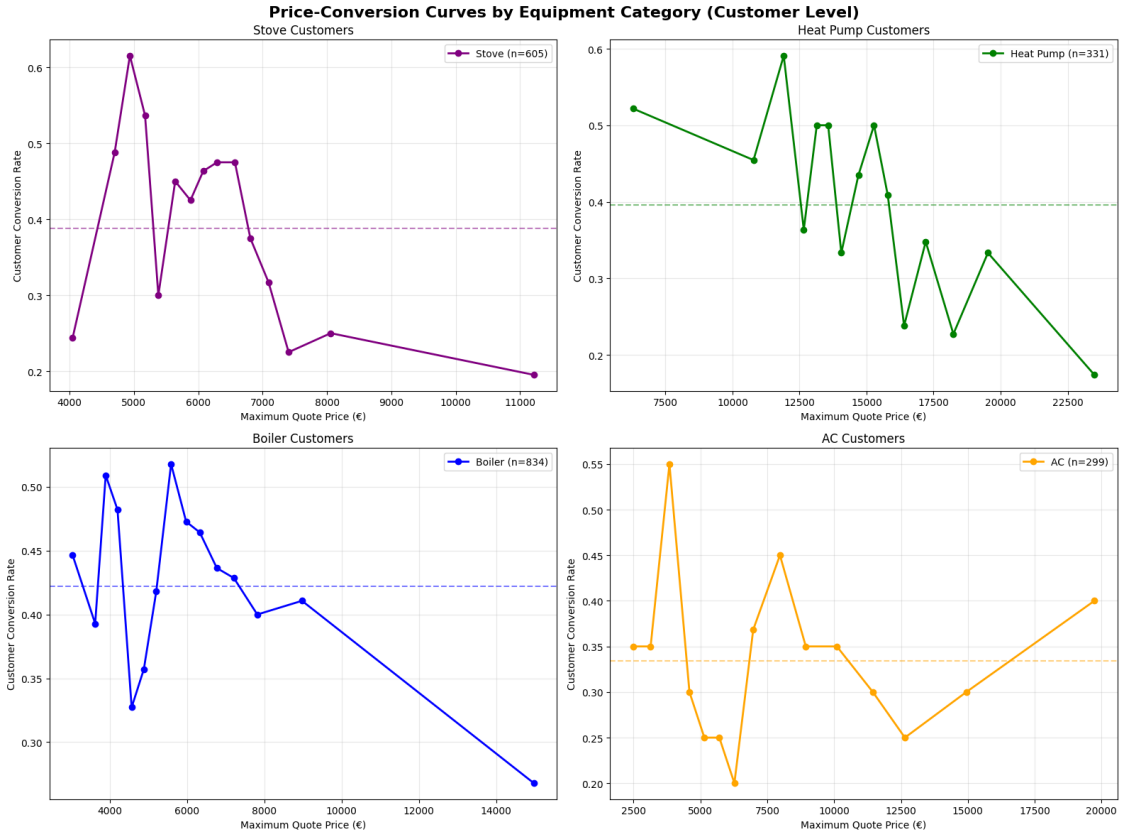


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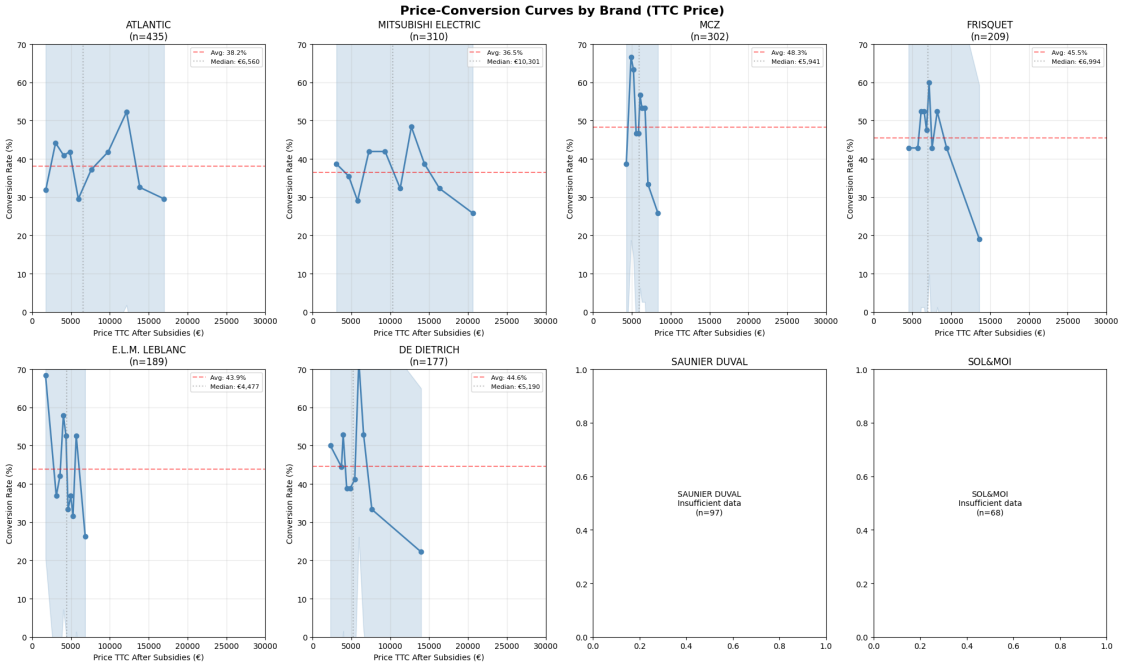
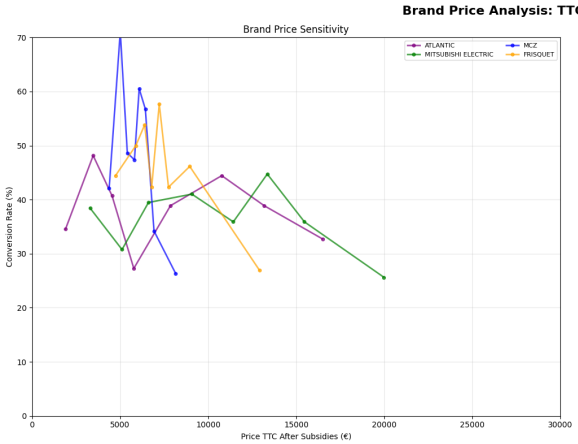
================================================================================
Conversion by Brand Price
================================================================================

Analyzing 2,414 customers
Price range: €1,312 to €26,385
Median price: €6,237
Top 8 brands: ['ATLANTIC', 'MITSUBISHI ELECTRIC', 'MCZ', 'FRISQUET', 'E.L.M.
LEBLANC', 'DE DIETRICH', 'SAUNIER DUVAL', 'SOL&MOI']
```




```
Analyzing 2,414 customers
Price range: €1,312 to €26,385
Median price: €6,237
```



Brand Performance Summary

Brand	Customers	Avg Conv	Avg Price	Best Price Range	Best Conv
ATLANTIC	435	38.2%	€7,999	€8,617-12,971	47.1%
MITSUBISHI ELECTRIC	310	36.5%	€10,542	€11,883-15,364	43.5%
MCZ	302	48.3%	€6,026	€5,009-5,738	55.0%
FRISQUET	209	45.5%	€7,568	€6,650-7,352	53.7%
E.L.M. LEBLANC	189	43.9%	€4,412	€1,347-3,410	52.6%
DE DIETRICH	177	44.6%	€5,881	€5,707-6,974	62.9%

```
================================================================================
TTC Price-conversion curves: 4 equipment types, multiple brands per graph
================================================================================

Analyzing 2,414 customers
```

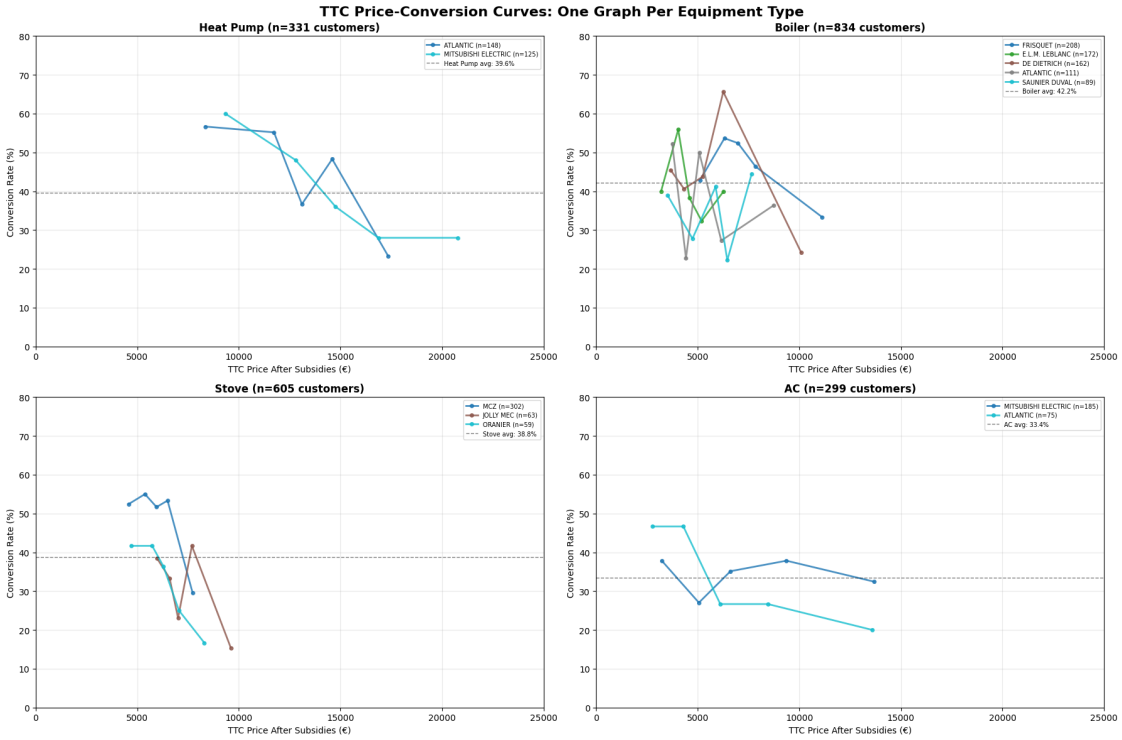
```
Price range: €1,312 to €26,385
Median price: €6,237

Heat Pump - Top brands: ['ATLANTIC', 'MITSUBISHI ELECTRIC']

Boiler - Top brands: ['FRISQUET', 'E.L.M. LEBLANC', 'DE DIETRICH', 'ATLANTIC',
'SAUNIER DUVAL']

Stove - Top brands: ['MCZ', 'JOLLY MEC', 'ORANIER']

AC - Top brands: ['MITSUBISHI ELECTRIC', 'ATLANTIC']
```

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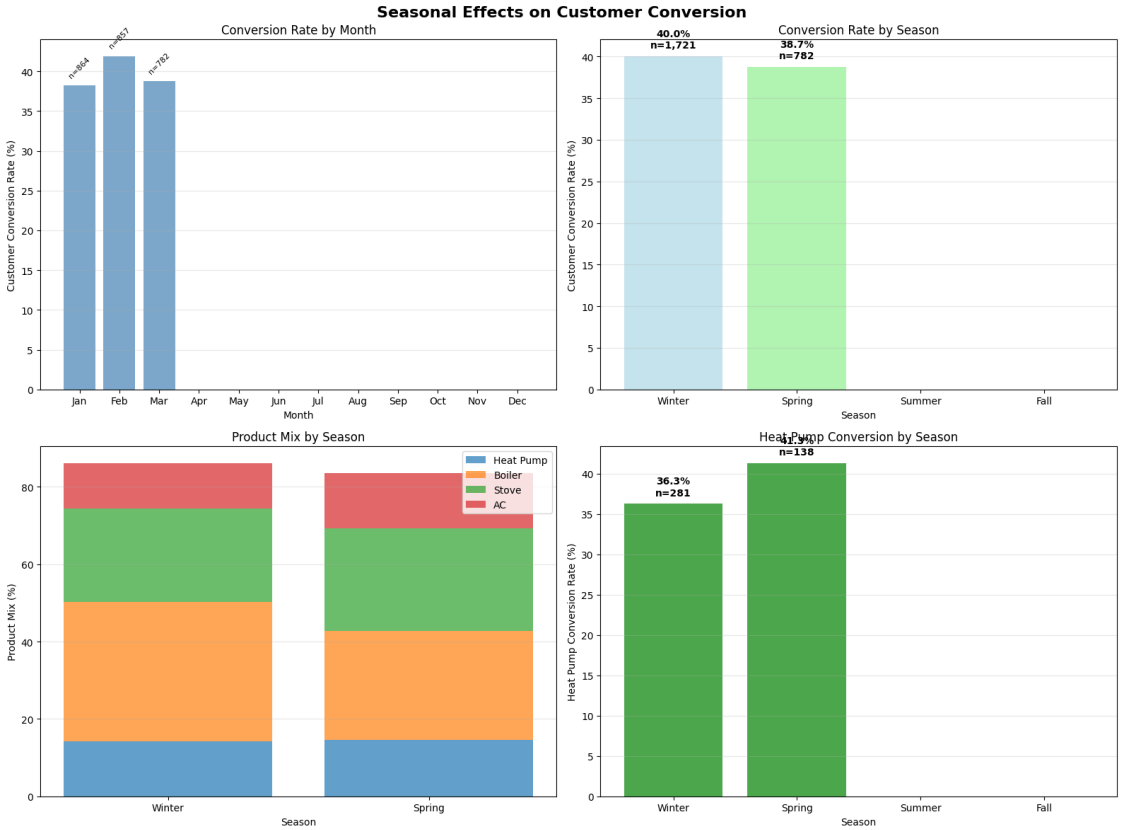
==========================================================================
Seasonal Effects on Customer Conversion
==========================================================================
```




```
================================================================================
SEASONAL STATISTICS
================================================================================

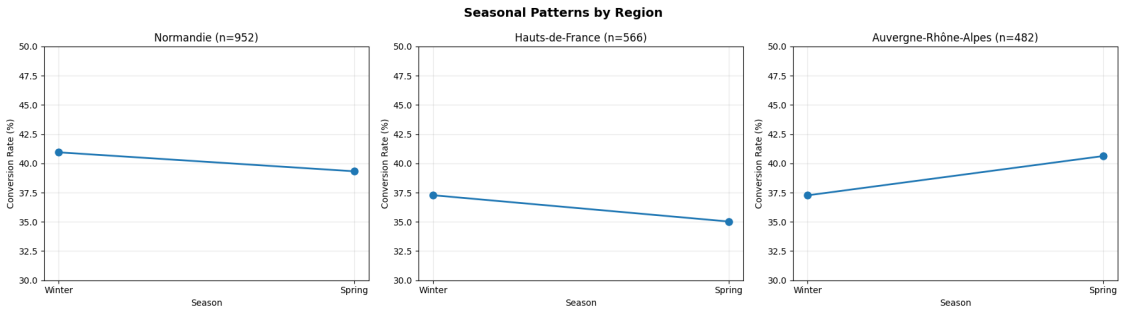
Conversion by season:
  Winter: 40.0% (n=1,721)
  Spring: 38.7% (n=782)

Seasonality chi-square p-value: 0.5710
Statistically significant: NO

================================================================================
Regional Seasonal Pattern
================================================================================

Analyzing 2,414 customers
Price range: €1,312 to €26,385
Median price: €6,237

Top regions: ['Normandie', 'Hauts-de-France', 'Auvergne-Rhône-Alpes']
```

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================================================================================
Year-over-year trends
================================================================================

Analyzing 2,414 customers
Price range: €1,312 to €26,385
Median price: €6,237

Quarterly conversion rates:
                  mean  count
year_quarter
2025Q1        0.397266   2414
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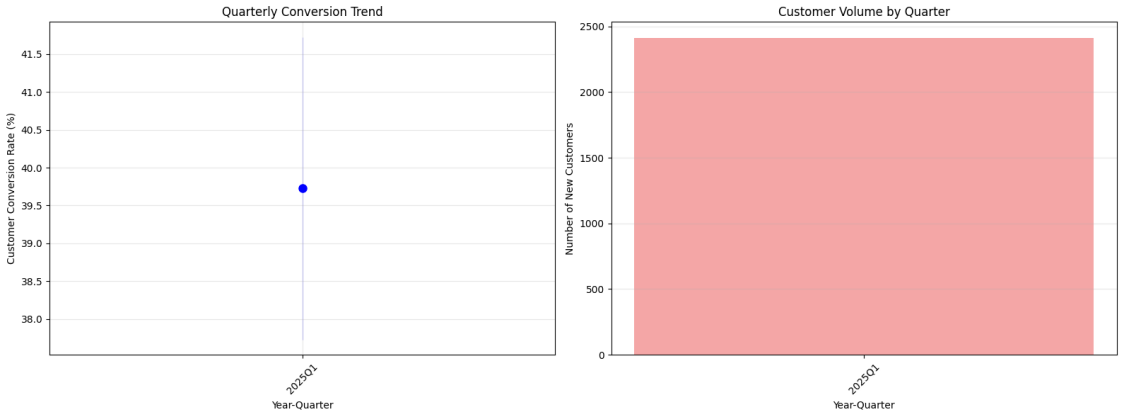


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================================================================================
Best and Worst Months Report
================================================================================

Analyzing 2,414 customers
Price range: €1,312 to €26,385
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```
Median price: €6,237

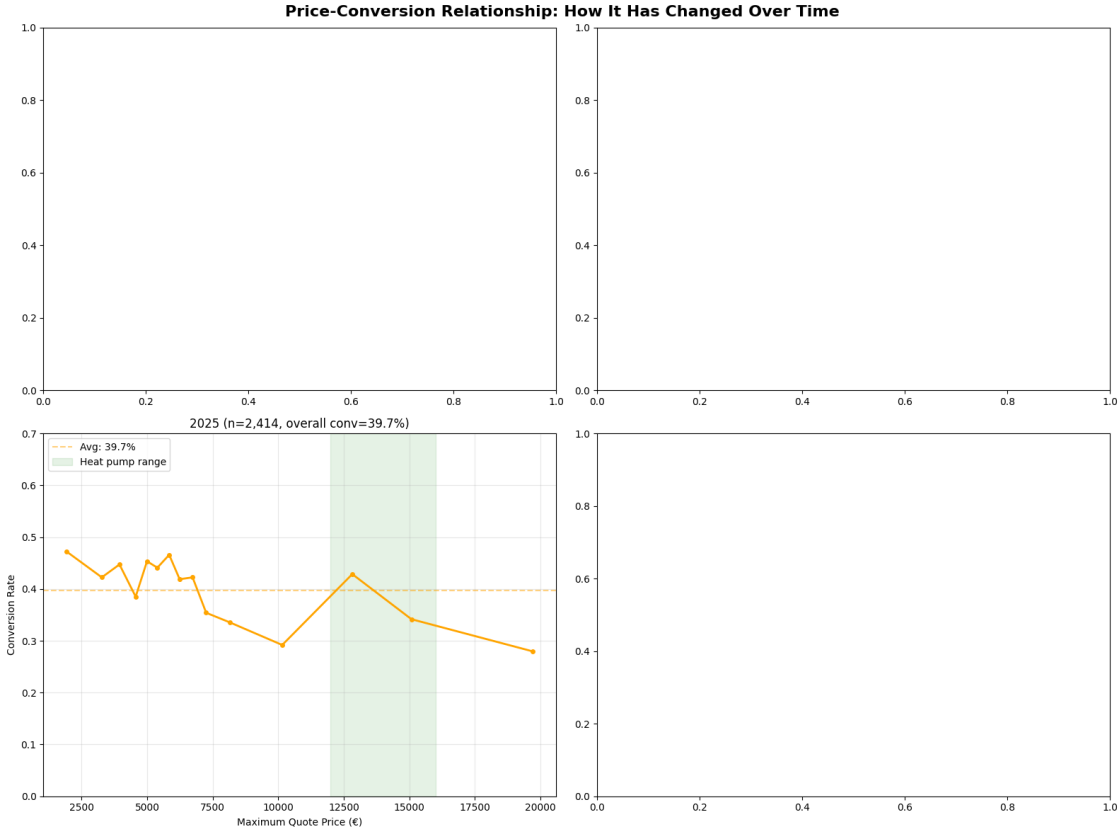
  Only one suspension period found: [False]
Skipping suspension comparison

Best month: 2 (42.3%, n=823)

Best season: Winter (40.2%)
Worst season: Spring (38.6%)

Heat pump conversion by season:
  Winter: 36.9%
  Spring: 43.2%

Analyzing 2,414 customers
Price range: €1,312 to €26,385
Median price: €6,237
```

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Analyzing 2,414 customers
Price range: €1,312 to €26,385
Median price: €6,237
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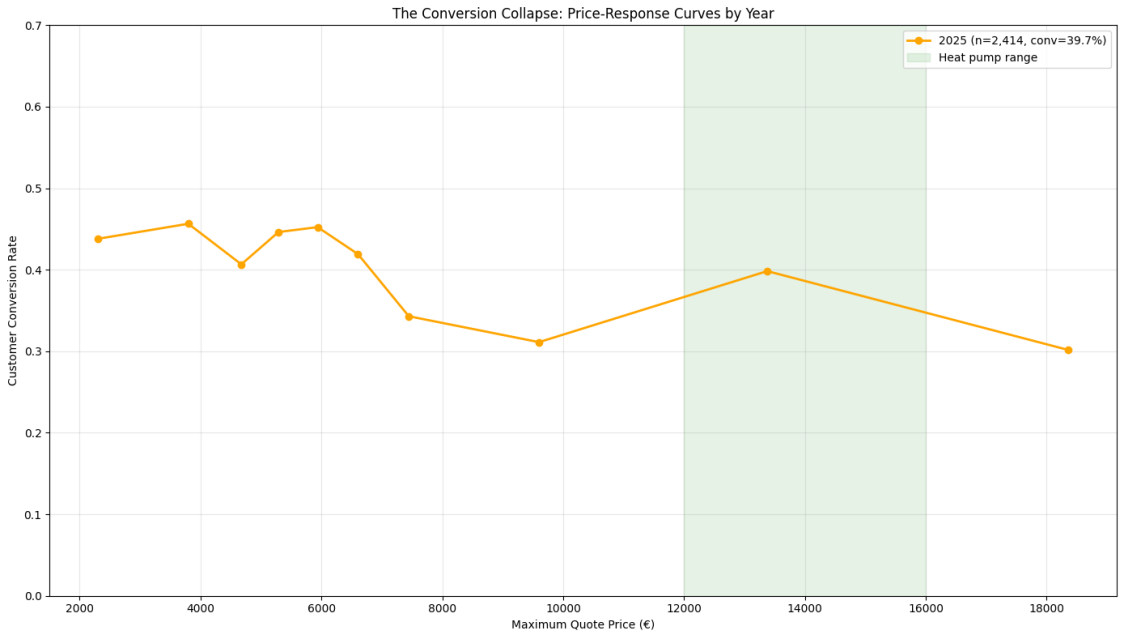


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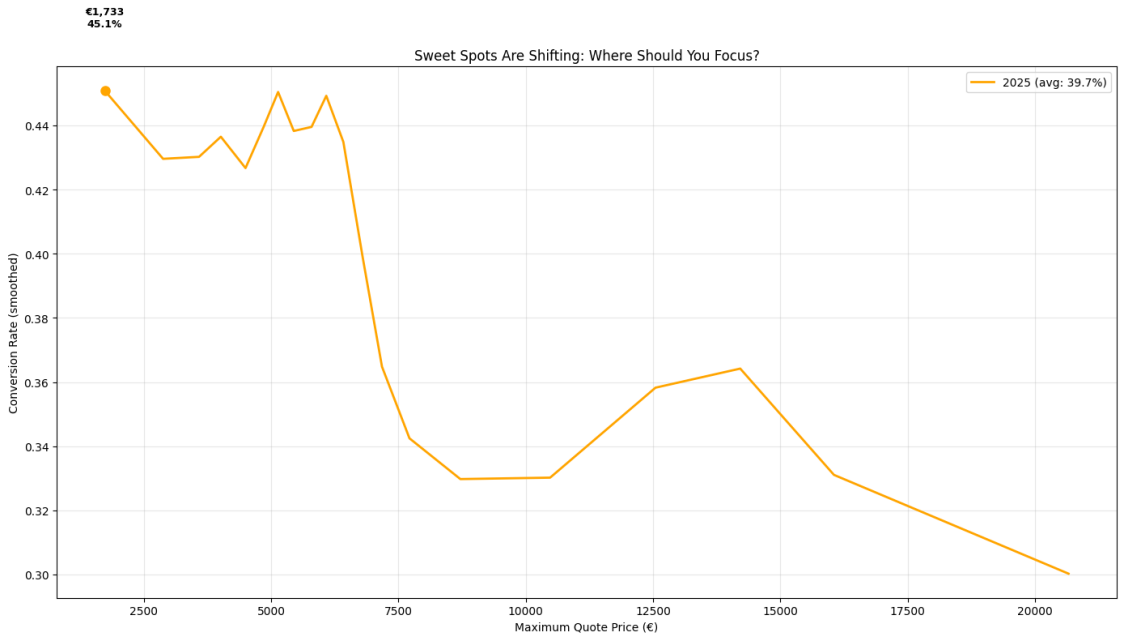
================================================================================
Sweet Spot Analysis Over Time
================================================================================

Analyzing 2,414 customers
Price range: €1,312 to €26,385
Median price: €6,237

Sweet spots by year (price points with conversion > avg + 3%):

2025 (overall avg: 39.7%):
  €1,733: 47.9% (n=121)
  €4,011: 50.4% (n=121)
  €4,857: 45.8% (n=120)
  €5,136: 48.8% (n=121)
  €5,792: 43.3% (n=120)
  €6,082: 47.1% (n=121)
  €6,417: 45.5% (n=121)
```

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================================================================================
Administrative Uncertainty Effect
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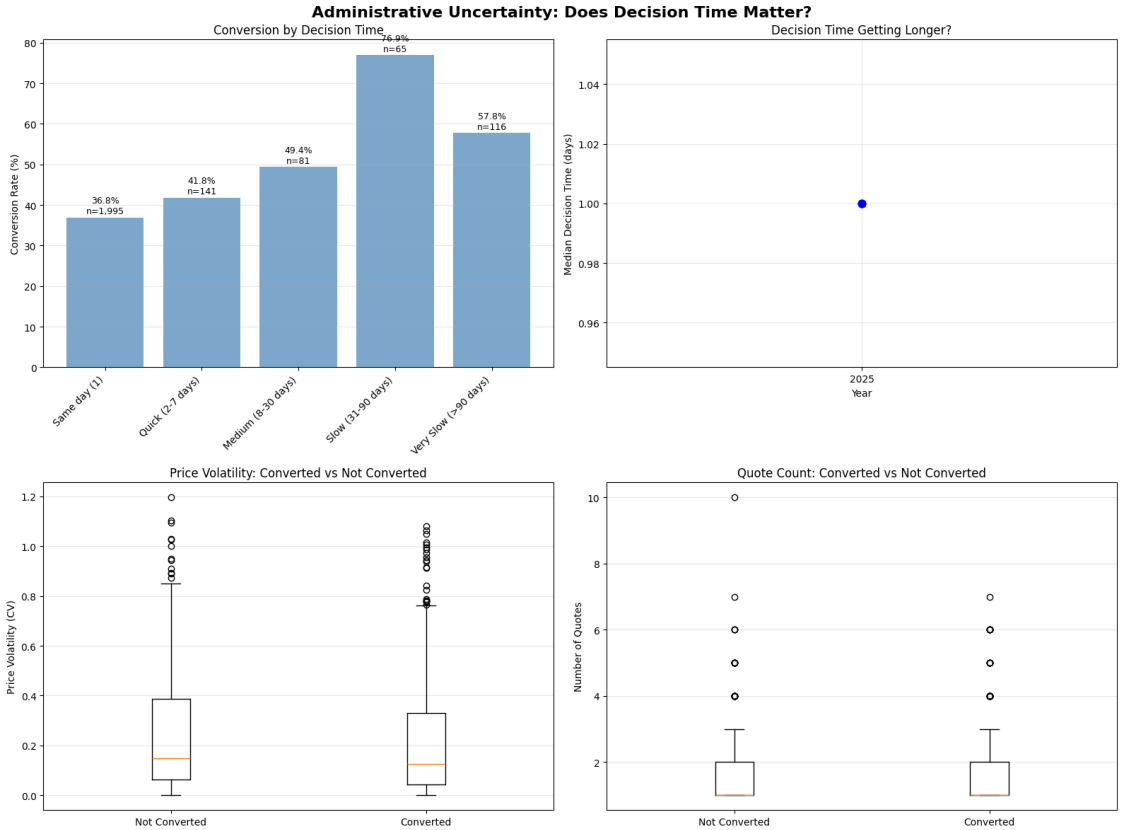
Analyzing 2,414 customers
Price range: €1,312 to €26,385
Median price: €6,237

Decision time distribution:
count    2414.000000
mean       15.908865
std        57.837648
min         1.000000
25%         1.000000
50%         1.000000
75%         1.000000
max       429.000000
Name: decision_days, dtype: float64

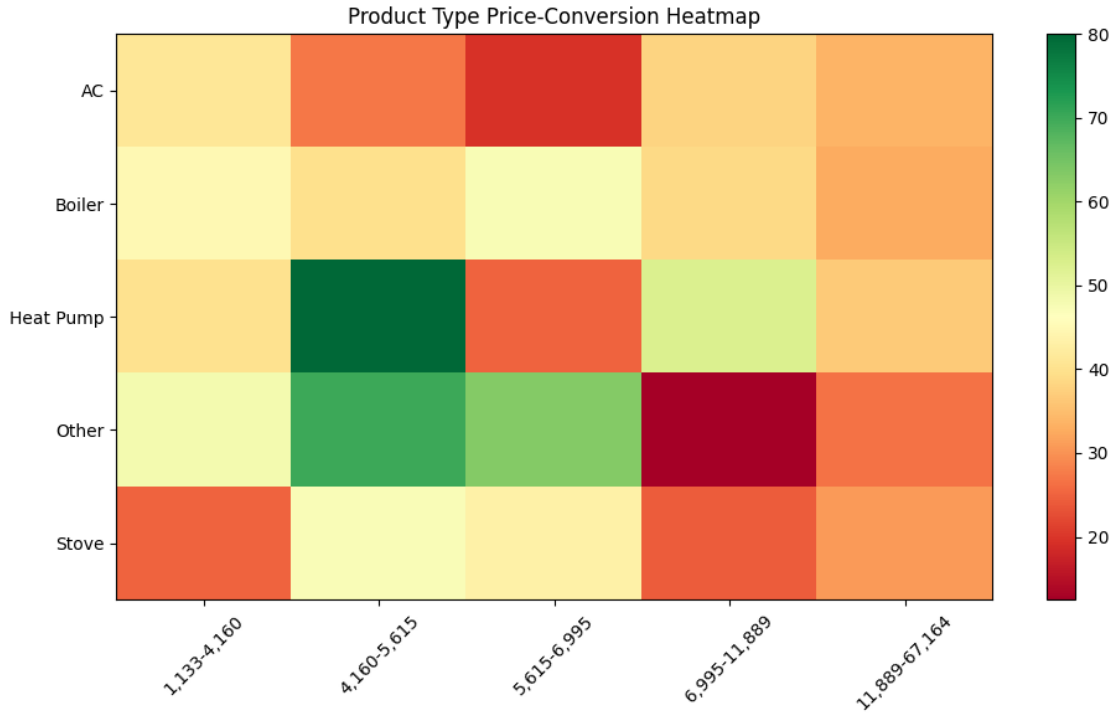
Customers with 1-day decision: 1,995 (82.6%)
```




```
================================================================================
Product Type - Price Conversion Heatmap
================================================================================
Simple pivot table:
price_bin_simple         (1133.359, 4160.648]  (4160.648, 5615.674]  \
main_equipment_category
AC                                  40.983607             27.083333
Boiler                              44.845361             39.737991
Heat Pump                           40.000000             80.000000
Other                               48.325359             70.000000
Stove                               25.000000             47.120419

price_bin_simple         (5615.674, 6995.726]  (6995.726, 11889.604]  \
main_equipment_category
AC                                  19.512195              37.931034
Boiler                              47.368421              38.797814
Heat Pump                           25.000000              52.380952
Other                               63.157895              12.500000
Stove                               43.277311              24.113475
```

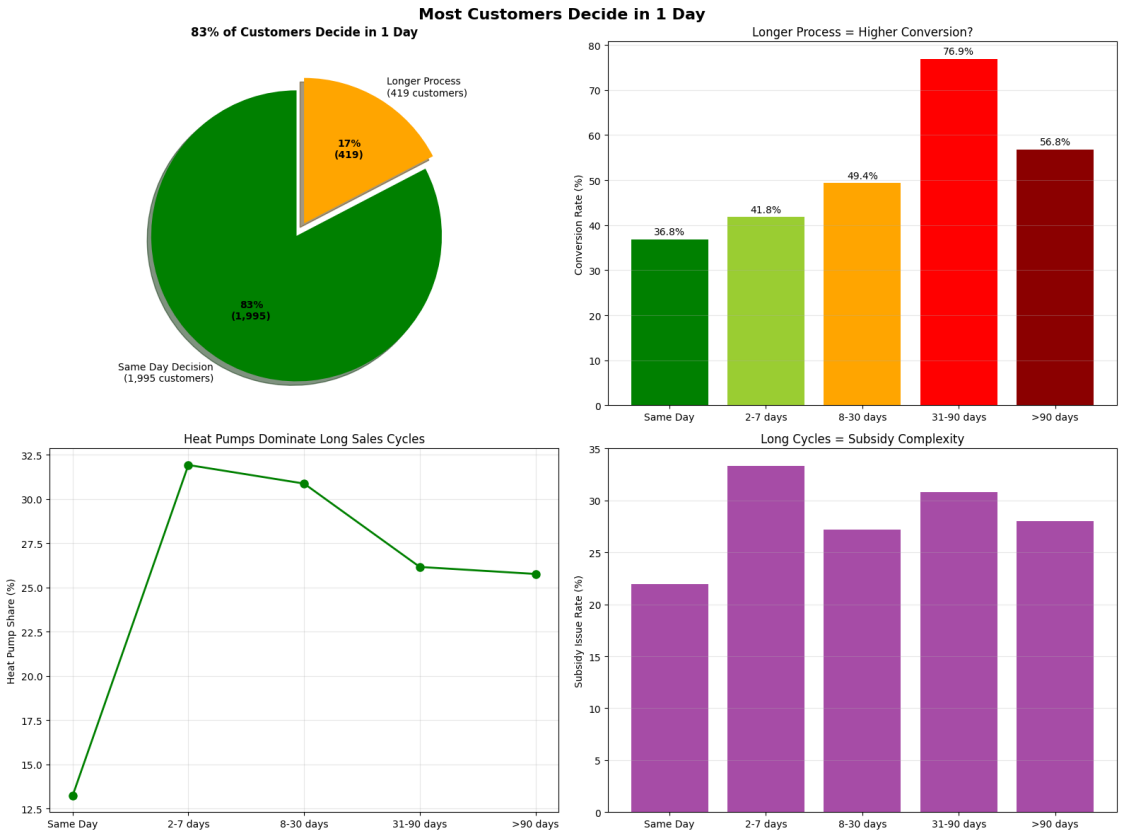
```
price_bin_simple         (11889.604, 67164.57]
main_equipment_category
AC                                   33.783784
Boiler                               32.558140
Heat Pump                            36.619718
Other                                26.582278
Stove                                30.769231
```




```
================================================================================
CHECKING MAIN PRODUCT FAMILY
================================================================================
main_product_family
Chaudière                          858
Poêle                              612
Pompe à chaleur                    337
Climatisation                      325
ECS : Chauffe-eau ou adoucisseur   157
Photovoltaïque                      73
Emetteur de chauffage  ou chappe    39
Unknown                             33
Autres                              24
Appareil hybride                    19
Accessoire de pose                  14
```

```
Produit VMC                          9
Plomberie Sanitaire                  3
Name: count, dtype: int64

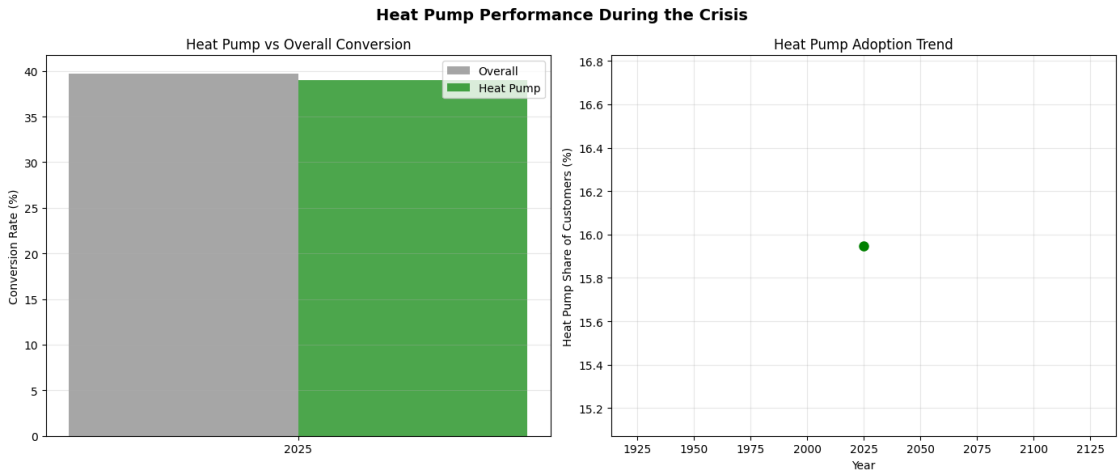
Analyzing 2,414 customers
Price range: €1,312 to €26,385
Median price: €6,237
```




```
================================================================================
Heat Pump Performance Over Time
================================================================================

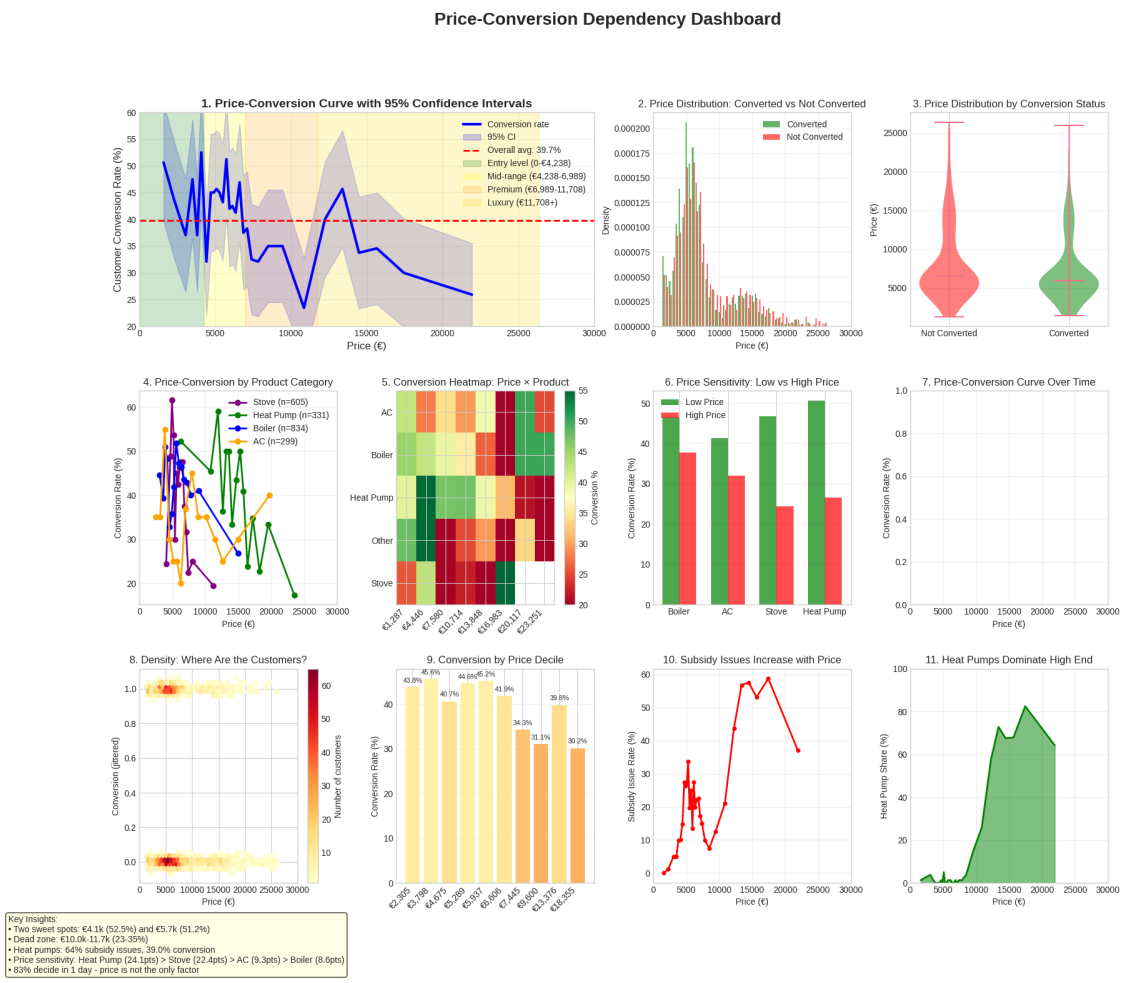
Analyzing 2,414 customers
Price range: €1,312 to €26,385
Median price: €6,237

Heat pump customers by year:
        conversion_rate  customer_count  avg_price  subsidy_issue_rate
period
2025               0.39             385  14893.519               0.636
```

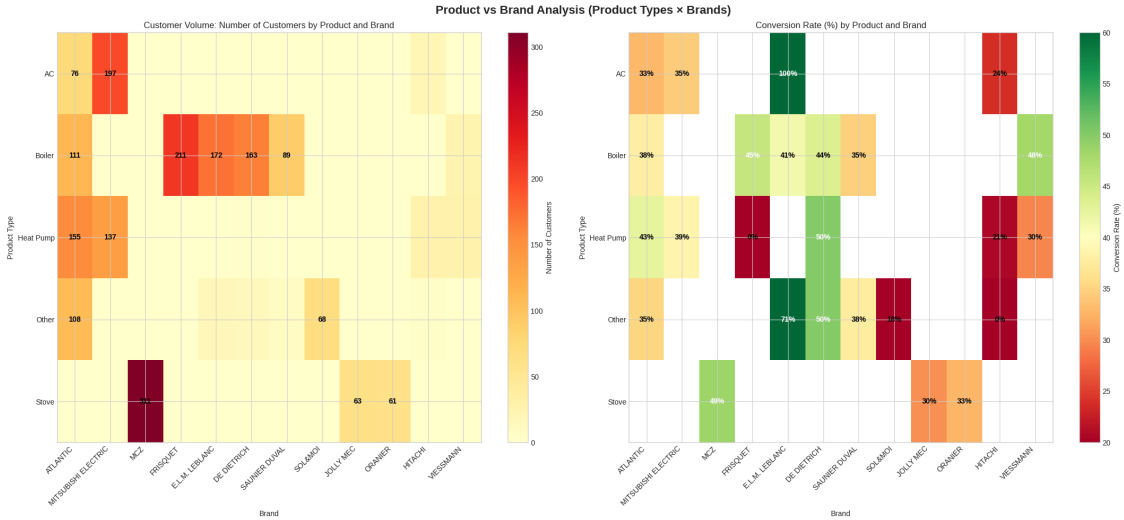



```
Analyzing 2,414 customers
Price range: €1,312 to €26,385
Median price: €6,237

================================================================================
DYNAMIC INSIGHTS CALCULATED:
================================================================================
Sweet spots: €4.1k (52.5%) and €5.7k (51.2%)
Dead zone: €10.0k-11.7k (23-35%)
Heat pumps: 64% subsidy issues, 39.0% conversion
Price sensitivity: Heat Pump (24.1pts) > Stove (22.4pts) > AC (9.3pts) > Boiler
(8.6pts)
Same day decisions: 83%
```



Actual product categories in data: ['AC', 'Boiler', 'Heat Pump', 'Other', 'Stove']



```
================================================================================
PRODUCT × BRAND ANALYSIS
================================================================================

  CUSTOMER VOLUME (Number of Customers):
--------------------------------------------------------------------------------
main_brand                ATLANTIC  MITSUBISHI ELECTRIC  MCZ  FRISQUET  E.L.M.
LEBLANC  DE DIETRICH  SAUNIER DUVAL  SOL&MOI  JOLLY MEC  ORANIER  HITACHI
VIESSMANN
main_equipment_category
AC                            76                  197    0         0
1            0              0        0          0        0       21          0
Boiler                       111                    0    0       211
172          163             89        0          0        0        0         25
Heat Pump                    155                  137    0         1
0            2              0        0          0        0       29         27
Other                        108                    0    0         0
17           14              8       68          0        0        4          0
Stove                          0                    0  311         0
0            0              0        0         63       61        0          0

================================================================================
  CONVERSION RATES (%):
--------------------------------------------------------------------------------
main_brand                ATLANTIC  MITSUBISHI ELECTRIC   MCZ  FRISQUET  E.L.M.
LEBLANC  DE DIETRICH  SAUNIER DUVAL  SOL&MOI  JOLLY MEC  ORANIER  HITACHI
VIESSMANN
main_equipment_category
AC                           32.9                 34.5   NaN       NaN
100.0          NaN            NaN      NaN        NaN      NaN     23.8
NaN
Boiler                       37.8                  NaN   NaN      45.5
41.3         43.6           34.8      NaN        NaN      NaN      NaN
48.0
Heat Pump                    42.6                 38.7   NaN       0.0
NaN         50.0            NaN      NaN        NaN      NaN     20.7       29.6
Other                        35.2                  NaN   NaN       NaN
70.6         50.0           37.5     17.6        NaN      NaN      0.0
NaN
Stove                         NaN                  NaN  48.6       NaN
NaN          NaN            NaN      NaN       30.2     32.8      NaN        NaN

================================================================================
  BEST BRAND BY PRODUCT TYPE
```

```
--------------------------------------------------------------------------------

AC:
  Best brand: E.L.M. LEBLANC
  Conversion: 100.0%
  Customers: 1
  Top 3 brands:
    1. E.L.M. LEBLANC: 100.0% (n=1)
    2. MITSUBISHI ELECTRIC: 34.5% (n=197)
    3. ATLANTIC: 32.9% (n=76)

Boiler:
  Best brand: VIESSMANN
  Conversion: 48.0%
  Customers: 25
  Top 3 brands:
    1. VIESSMANN: 48.0% (n=25)
    2. FRISQUET: 45.5% (n=211)
    3. DE DIETRICH: 43.6% (n=163)

Heat Pump:
  Best brand: DE DIETRICH
  Conversion: 50.0%
  Customers: 2
  Top 3 brands:
    1. DE DIETRICH: 50.0% (n=2)
    2. ATLANTIC: 42.6% (n=155)
    3. MITSUBISHI ELECTRIC: 38.7% (n=137)

Other:
  Best brand: E.L.M. LEBLANC
  Conversion: 70.6%
  Customers: 17
  Top 3 brands:
    1. E.L.M. LEBLANC: 70.6% (n=17)
    2. DE DIETRICH: 50.0% (n=14)
    3. SAUNIER DUVAL: 37.5% (n=8)

Stove:
  Best brand: MCZ
  Conversion: 48.6%
  Customers: 311
  Top 3 brands:
    1. MCZ: 48.6% (n=311)
    2. ORANIER: 32.8% (n=61)
    3. JOLLY MEC: 30.2% (n=63)

================================================================================
```

```
  BEST PRODUCT BY BRAND
--------------------------------------------------------------------------------

ATLANTIC:
  Best product: Heat Pump
  Conversion: 42.6%
  Customers: 155
  All products:
    Heat Pump: 42.6% (n=155)
    Boiler: 37.8% (n=111)
    Other: 35.2% (n=108)
    AC: 32.9% (n=76)

MITSUBISHI ELECTRIC:
  Best product: Heat Pump
  Conversion: 38.7%
  Customers: 137
  All products:
    Heat Pump: 38.7% (n=137)
    AC: 34.5% (n=197)

MCZ:
  Best product: Stove
  Conversion: 48.6%
  Customers: 311
  All products:
    Stove: 48.6% (n=311)

FRISQUET:
  Best product: Boiler
  Conversion: 45.5%
  Customers: 211
  All products:
    Boiler: 45.5% (n=211)
    Heat Pump: 0.0% (n=1)

E.L.M. LEBLANC:
  Best product: AC
  Conversion: 100.0%
  Customers: 1
  All products:
    AC: 100.0% (n=1)
    Other: 70.6% (n=17)
    Boiler: 41.3% (n=172)

DE DIETRICH:
  Best product: Heat Pump
  Conversion: 50.0%
```

```
  Customers: 2
  All products:
    Heat Pump: 50.0% (n=2)
    Other: 50.0% (n=14)
    Boiler: 43.6% (n=163)

SAUNIER DUVAL:
  Best product: Other
  Conversion: 37.5%
  Customers: 8
  All products:
    Other: 37.5% (n=8)
    Boiler: 34.8% (n=89)

SOL&MOI:
  Best product: Other
  Conversion: 17.6%
  Customers: 68
  All products:
    Other: 17.6% (n=68)

================================================================================
 STRATEGIC INSIGHTS
================================================================================

Brand Dominance by Product Category:

AC:
  Volume leader: MITSUBISHI ELECTRIC (197 customers, 66.8% share)
  Conversion leader: E.L.M. LEBLANC (100.0%)

Boiler:
  Volume leader: FRISQUET (211 customers, 27.4% share)
  Conversion leader: VIESSMANN (48.0%)

Heat Pump:
  Volume leader: ATLANTIC (155 customers, 44.2% share)
  Conversion leader: DE DIETRICH (50.0%)

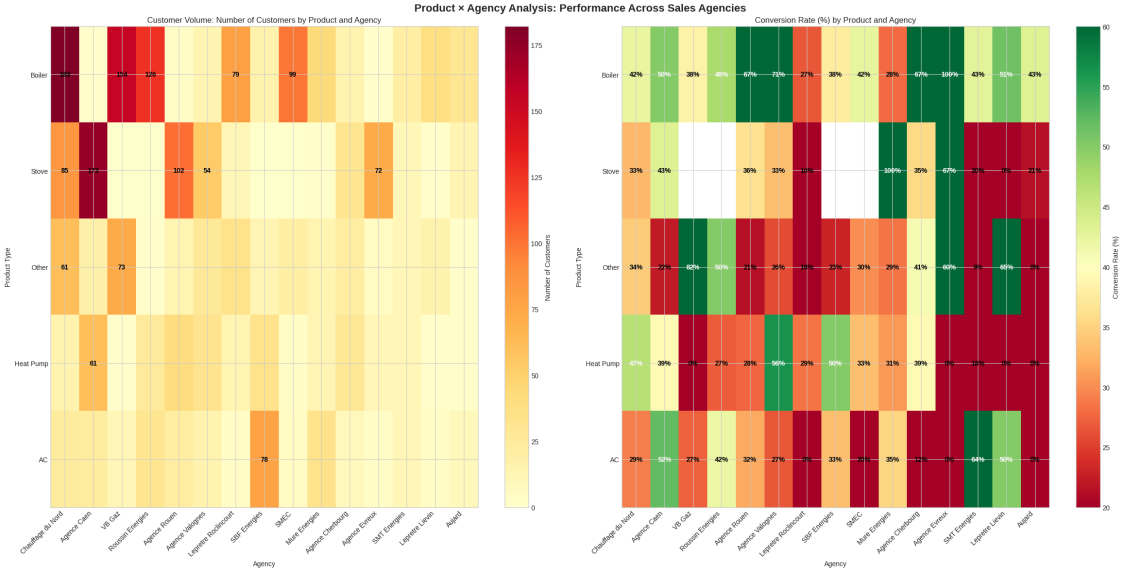
Other:
  Volume leader: ATLANTIC (108 customers, 49.3% share)
  Conversion leader: E.L.M. LEBLANC (70.6%)

Stove:
  Volume leader: MCZ (311 customers, 71.5% share)
  Conversion leader: MCZ (48.6%)

Actual product categories in data: ['Boiler', 'Stove', 'Other', 'Heat Pump',
```

```
'AC']

Top 15 agencies by customer volume:
  Chauffage du Nord: 367 customers, 38.1% conversion
  Agence Caen: 277 customers, 41.9% conversion
  VB Gaz: 239 customers, 51.0% conversion
  Roussin Energies: 185 customers, 43.8% conversion
  Agence Rouen: 181 customers, 33.1% conversion
  Agence Valognes: 142 customers, 40.1% conversion
  Lepretre Roclincourt: 141 customers, 22.7% conversion
  SBF Energies: 141 customers, 36.9% conversion
  SMEC: 127 customers, 39.4% conversion
  Mure Energies: 106 customers, 32.1% conversion
  Agence Cherbourg: 104 customers, 40.4% conversion
  Agence Evreux: 93 customers, 55.9% conversion
  SMT Energies: 64 customers, 32.8% conversion
  Lepretre Lievin: 63 customers, 52.4% conversion
  Aujard: 57 customers, 28.1% conversion
```




```
================================================================================
PRODUCT × AGENCY ANALYSIS
================================================================================

  CUSTOMER VOLUME BY PRODUCT AND AGENCY:
--------------------------------------------------------------------------------
--------------------
main_agency                Chauffage du Nord  Agence Caen  VB Gaz  Roussin
Energies  Agence Rouen  Agence Valognes  Lepretre Roclincourt  SBF Energies
```

```
SMEC  Mure Energies  Agence Cherbourg  Agence Evreux  SMT Energies  Lepretre
Lievin  Aujard
main_equipment_category
Boiler                                  182            2     154
126             3               14                     79            16    99
43                15              1            21                39      30
Stove                                    85          173       0
0           102               54                     10             0     0
2                31             72            10                 1      14
Other                                    61           18      73
2            14               27                     34            13    20
14                22              5            11                17       2
Heat Pump                                15           61       1
26            40               32                     14            34     3
13                28             11            11                 2       2
AC                                       24           23      11
31            22               15                      4            78     5
34                 8              4            11                 4       9

================================================================================
 CONVERSION RATES (%) BY PRODUCT AND AGENCY:
--------------------------------------------------------------------------------
--------------------
main_agency              Chauffage du Nord  Agence Caen  VB Gaz  Roussin
Energies  Agence Rouen  Agence Valognes  Lepretre Roclincourt  SBF Energies
SMEC  Mure Energies  Agence Cherbourg  Agence Evreux  SMT Energies  Lepretre
Lievin  Aujard
main_equipment_category
Boiler                                 42.3        50.0    38.3
47.6          66.7             71.4                  26.6          37.5  42.4
27.9              66.7          100.0          42.9             51.3    43.3
Stove                                  32.9        43.4     NaN
NaN          36.3             33.3                  10.0           NaN   NaN
100.0              35.5           66.7          20.0              0.0    21.4
Other                                  34.4        22.2    82.2
50.0          21.4             25.9                  17.6          23.1  30.0
28.6              40.9           60.0           9.1             64.7     0.0
Heat Pump                              46.7        39.3     0.0
26.9          27.5             56.2                  28.6          50.0  33.3
30.8              39.3            0.0          18.2              0.0     0.0
AC                                     29.2        52.2    27.3
41.9          31.8             26.7                   0.0          33.3  20.0
35.3              12.5            0.0          63.6             50.0     0.0

================================================================================
 BEST AGENCY BY PRODUCT TYPE
--------------------------------------------------------------------------------
--------------------
```

```
Boiler:
  Best agency: Agence Evreux
  Conversion: 100.0%
  Customers: 1
  Top 3 agencies:
    1. Agence Evreux: 100.0% (n=1)
    2. Agence Valognes: 71.4% (n=14)
    3. Agence Rouen: 66.7% (n=3)

Stove:
  Best agency: Mure Energies
  Conversion: 100.0%
  Customers: 2
  Top 3 agencies:
    1. Mure Energies: 100.0% (n=2)
    2. Agence Evreux: 66.7% (n=72)
    3. Agence Caen: 43.4% (n=173)

Other:
  Best agency: VB Gaz
  Conversion: 82.2%
  Customers: 73
  Top 3 agencies:
    1. VB Gaz: 82.2% (n=73)
    2. Lepretre Lievin: 64.7% (n=17)
    3. Agence Evreux: 60.0% (n=5)

Heat Pump:
  Best agency: Agence Valognes
  Conversion: 56.2%
  Customers: 32
  Top 3 agencies:
    1. Agence Valognes: 56.2% (n=32)
    2. SBF Energies: 50.0% (n=34)
    3. Chauffage du Nord: 46.7% (n=15)

AC:
  Best agency: SMT Energies
  Conversion: 63.6%
  Customers: 11
  Top 3 agencies:
    1. SMT Energies: 63.6% (n=11)
    2. Agence Caen: 52.2% (n=23)
    3. Lepretre Lievin: 50.0% (n=4)

================================================================================
 BEST PRODUCT BY AGENCY
```

```
--------------------------------------------------------------------------------
--------------------

Chauffage du Nord:
  Best product: Heat Pump
  Conversion: 46.7%
  Customers: 15
  All products:
    Heat Pump: 46.7% (n=15)
    Boiler: 42.3% (n=182)
    Other: 34.4% (n=61)
    Stove: 32.9% (n=85)
    AC: 29.2% (n=24)

Agence Caen:
  Best product: AC
  Conversion: 52.2%
  Customers: 23
  All products:
    AC: 52.2% (n=23)
    Boiler: 50.0% (n=2)
    Stove: 43.4% (n=173)
    Heat Pump: 39.3% (n=61)
    Other: 22.2% (n=18)

VB Gaz:
  Best product: Other
  Conversion: 82.2%
  Customers: 73
  All products:
    Other: 82.2% (n=73)
    Boiler: 38.3% (n=154)
    AC: 27.3% (n=11)
    Heat Pump: 0.0% (n=1)

Roussin Energies:
  Best product: Other
  Conversion: 50.0%
  Customers: 2
  All products:
    Other: 50.0% (n=2)
    Boiler: 47.6% (n=126)
    AC: 41.9% (n=31)
    Heat Pump: 26.9% (n=26)

Agence Rouen:
  Best product: Boiler
  Conversion: 66.7%
```

```
  Customers: 3
  All products:
    Boiler: 66.7% (n=3)
    Stove: 36.3% (n=102)
    AC: 31.8% (n=22)
    Heat Pump: 27.5% (n=40)
    Other: 21.4% (n=14)

Agence Valognes:
  Best product: Boiler
  Conversion: 71.4%
  Customers: 14
  All products:
    Boiler: 71.4% (n=14)
    Heat Pump: 56.2% (n=32)
    Stove: 33.3% (n=54)
    AC: 26.7% (n=15)
    Other: 25.9% (n=27)

Lepretre Roclincourt:
  Best product: Heat Pump
  Conversion: 28.6%
  Customers: 14
  All products:
    Heat Pump: 28.6% (n=14)
    Boiler: 26.6% (n=79)
    Other: 17.6% (n=34)
    Stove: 10.0% (n=10)
    AC: 0.0% (n=4)

SBF Energies:
  Best product: Heat Pump
  Conversion: 50.0%
  Customers: 34
  All products:
    Heat Pump: 50.0% (n=34)
    Boiler: 37.5% (n=16)
    AC: 33.3% (n=78)
    Other: 23.1% (n=13)

SMEC:
  Best product: Boiler
  Conversion: 42.4%
  Customers: 99
  All products:
    Boiler: 42.4% (n=99)
    Heat Pump: 33.3% (n=3)
    Other: 30.0% (n=20)
```

```
    AC: 20.0% (n=5)

Mure Energies:
  Best product: Stove
  Conversion: 100.0%
  Customers: 2
  All products:
    Stove: 100.0% (n=2)
    AC: 35.3% (n=34)
    Heat Pump: 30.8% (n=13)
    Other: 28.6% (n=14)
    Boiler: 27.9% (n=43)

================================================================================
 AGENCY PERFORMANCE SUMMARY
================================================================================

Top 15 agencies by volume:
                      conversion_rate  customer_count  heat_pump_pct
subsidy_issue_pct  avg_price
main_agency
Chauffage du Nord                38.1             367            5.7
26.7   6051.773
Agence Caen                      41.9             277           23.5
36.5   8162.859
VB Gaz                           51.0             239            1.3
0.0   4484.461
Roussin Energies                 43.8             185           17.3
10.3   7604.360
Agence Rouen                     33.1             181           24.9
48.1   9006.808
Agence Valognes                  40.1             142           27.5
37.3   9306.251
Lepretre Roclincourt             22.7             141           10.6
13.5   6708.743
SBF Energies                     36.9             141           29.8
19.9   9315.245
SMEC                             39.4             127            2.4
0.8   6062.636
Mure Energies                    32.1             106           14.2
15.1   6711.071
Agence Cherbourg                 40.4             104           31.7
35.6   8413.730
Agence Evreux                    55.9              93           14.0
47.3   7101.228
SMT Energies                     32.8              64           25.0
17.2   6936.117
Lepretre Lievin                  52.4              63            3.2
```

```
3.2   5981.427
Aujard                          28.1              57            3.5
1.8   8372.007

================================================================================
  STRATEGIC INSIGHTS
================================================================================

Agency Excellence by Product Category:

Boiler:
  Conversion leader: Agence Evreux (100.0%, n=1)

Stove:
  Conversion leader: Mure Energies (100.0%, n=2)

Other:
  Conversion leader: VB Gaz (82.2%, n=73)

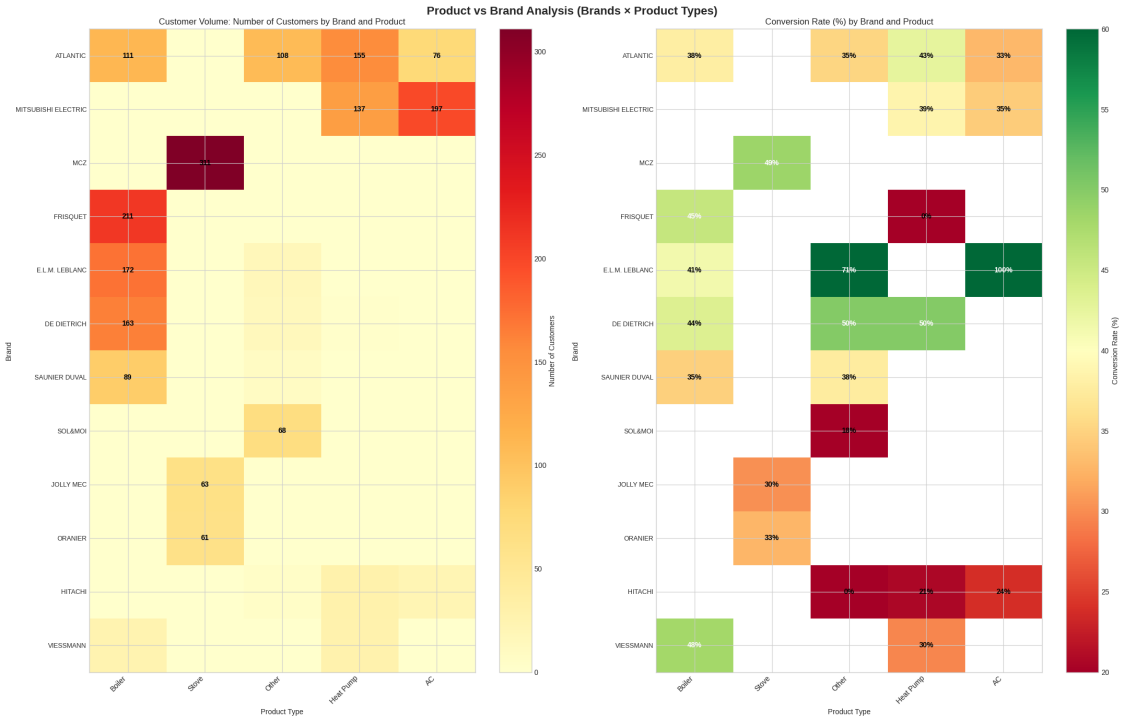
Heat Pump:
  Conversion leader: Agence Valognes (56.2%, n=32)

AC:
  Conversion leader: SMT Energies (63.6%, n=11)

================================================================================
  HEAT PUMP SPECIALISTS
================================================================================

Top 5 agencies for Heat Pump conversion:
  Agence Valognes: 56.2% (n=32)
  SBF Energies: 50.0% (n=34)
  Chauffage du Nord: 46.7% (n=15)
  Agence Caen: 39.3% (n=61)
  Agence Cherbourg: 39.3% (n=28)

Actual product categories in data: ['Boiler', 'Stove', 'Other', 'Heat Pump',
'AC']
```




```
================================================================================
BRAND × PRODUCT ANALYSIS (Swapped Axes)
================================================================================

  CUSTOMER VOLUME (Number of Customers):
--------------------------------------------------------------------------------
main_equipment_category  Boiler  Stove  Other  Heat Pump   AC
main_brand
ATLANTIC                   111      0    108        155   76
MITSUBISHI ELECTRIC          0      0      0        137  197
MCZ                          0    311      0          0    0
FRISQUET                   211      0      0          1    0
E.L.M. LEBLANC             172      0     17          0    1
DE DIETRICH                163      0     14          2    0
SAUNIER DUVAL               89      0      8          0    0
SOL&MOI                      0      0     68          0    0
JOLLY MEC                    0     63      0          0    0
ORANIER                      0     61      0          0    0
HITACHI                      0      0      4         29   21
VIESSMANN                   25      0      0         27    0

================================================================================
  CONVERSION RATES (%):
```

```
--------------------------------------------------------------------------------
main_equipment_category  Boiler  Stove  Other  Heat Pump     AC
main_brand
ATLANTIC                   37.8    NaN   35.2       42.6   32.9
MITSUBISHI ELECTRIC         NaN    NaN    NaN       38.7   34.5
MCZ                         NaN   48.6    NaN        NaN    NaN
FRISQUET                   45.5    NaN    NaN        0.0    NaN
E.L.M. LEBLANC             41.3    NaN   70.6        NaN  100.0
DE DIETRICH                43.6    NaN   50.0       50.0    NaN
SAUNIER DUVAL              34.8    NaN   37.5        NaN    NaN
SOL&MOI                     NaN    NaN   17.6        NaN    NaN
JOLLY MEC                   NaN   30.2    NaN        NaN    NaN
ORANIER                     NaN   32.8    NaN        NaN    NaN
HITACHI                     NaN    NaN    0.0       20.7   23.8
VIESSMANN                  48.0    NaN    NaN       29.6    NaN

================================================================================
  BEST PRODUCT BY BRAND
--------------------------------------------------------------------------------

ATLANTIC:
  Best product: Heat Pump
  Conversion: 42.6%
  Customers: 155
  Top 3 products:
    1. Heat Pump: 42.6% (n=155)
    2. Boiler: 37.8% (n=111)
    3. Other: 35.2% (n=108)

MITSUBISHI ELECTRIC:
  Best product: Heat Pump
  Conversion: 38.7%
  Customers: 137
  Top 3 products:
    1. Heat Pump: 38.7% (n=137)
    2. AC: 34.5% (n=197)

MCZ:
  Best product: Stove
  Conversion: 48.6%
  Customers: 311
  Top 3 products:
    1. Stove: 48.6% (n=311)

FRISQUET:
  Best product: Boiler
  Conversion: 45.5%
  Customers: 211
```

```
  Top 3 products:
    1. Boiler: 45.5% (n=211)
    2. Heat Pump: 0.0% (n=1)

E.L.M. LEBLANC:
  Best product: AC
  Conversion: 100.0%
  Customers: 1
  Top 3 products:
    1. AC: 100.0% (n=1)
    2. Other: 70.6% (n=17)
    3. Boiler: 41.3% (n=172)

DE DIETRICH:
  Best product: Other
  Conversion: 50.0%
  Customers: 14
  Top 3 products:
    1. Other: 50.0% (n=14)
    2. Heat Pump: 50.0% (n=2)
    3. Boiler: 43.6% (n=163)

SAUNIER DUVAL:
  Best product: Other
  Conversion: 37.5%
  Customers: 8
  Top 3 products:
    1. Other: 37.5% (n=8)
    2. Boiler: 34.8% (n=89)

SOL&MOI:
  Best product: Other
  Conversion: 17.6%
  Customers: 68
  Top 3 products:
    1. Other: 17.6% (n=68)

JOLLY MEC:
  Best product: Stove
  Conversion: 30.2%
  Customers: 63
  Top 3 products:
    1. Stove: 30.2% (n=63)

ORANIER:
  Best product: Stove
  Conversion: 32.8%
  Customers: 61
```

```
  Top 3 products:
    1. Stove: 32.8% (n=61)

================================================================================
 BEST BRAND BY PRODUCT TYPE
--------------------------------------------------------------------------------

Boiler:
  Best brand: VIESSMANN
  Conversion: 48.0%
  Customers: 25
  Top 5 brands:
    1. VIESSMANN: 48.0% (n=25)
    2. FRISQUET: 45.5% (n=211)
    3. DE DIETRICH: 43.6% (n=163)
    4. E.L.M. LEBLANC: 41.3% (n=172)
    5. ATLANTIC: 37.8% (n=111)

Stove:
  Best brand: MCZ
  Conversion: 48.6%
  Customers: 311
  Top 5 brands:
    1. MCZ: 48.6% (n=311)
    2. ORANIER: 32.8% (n=61)
    3. JOLLY MEC: 30.2% (n=63)

Other:
  Best brand: E.L.M. LEBLANC
  Conversion: 70.6%
  Customers: 17
  Top 5 brands:
    1. E.L.M. LEBLANC: 70.6% (n=17)
    2. DE DIETRICH: 50.0% (n=14)
    3. SAUNIER DUVAL: 37.5% (n=8)
    4. ATLANTIC: 35.2% (n=108)
    5. SOL&MOI: 17.6% (n=68)

Heat Pump:
  Best brand: DE DIETRICH
  Conversion: 50.0%
  Customers: 2
  Top 5 brands:
    1. DE DIETRICH: 50.0% (n=2)
    2. ATLANTIC: 42.6% (n=155)
    3. MITSUBISHI ELECTRIC: 38.7% (n=137)
    4. VIESSMANN: 29.6% (n=27)
    5. HITACHI: 20.7% (n=29)
```

```
AC:
  Best brand: E.L.M. LEBLANC
  Conversion: 100.0%
  Customers: 1
  Top 5 brands:
    1. E.L.M. LEBLANC: 100.0% (n=1)
    2. MITSUBISHI ELECTRIC: 34.5% (n=197)
    3. ATLANTIC: 32.9% (n=76)
    4. HITACHI: 23.8% (n=21)

================================================================================
 STRATEGIC INSIGHTS
================================================================================

Brand Performance by Product Category:

ATLANTIC:
  Heat Pump: 42.6% (n=155)
  Boiler: 37.8% (n=111)
  Other: 35.2% (n=108)

MITSUBISHI ELECTRIC:
  Heat Pump: 38.7% (n=137)
  AC: 34.5% (n=197)

MCZ:
  Stove: 48.6% (n=311)

FRISQUET:
  Boiler: 45.5% (n=211)
  Heat Pump: 0.0% (n=1)

E.L.M. LEBLANC:
  AC: 100.0% (n=1)
  Other: 70.6% (n=17)
  Boiler: 41.3% (n=172)

DE DIETRICH:
  Other: 50.0% (n=14)
  Heat Pump: 50.0% (n=2)
  Boiler: 43.6% (n=163)

SAUNIER DUVAL:
  Other: 37.5% (n=8)
  Boiler: 34.8% (n=89)

SOL&MOI:
```

```
  Other: 17.6% (n=68)

================================================================================
 MARKET LEADERS BY PRODUCT CATEGORY
================================================================================

Boiler:
  Volume leader: FRISQUET (211 customers, 27.4% share)
  Conversion leader: VIESSMANN (48.0%)

Stove:
  Volume leader: MCZ (311 customers, 71.5% share)
  Conversion leader: MCZ (48.6%)

Other:
  Volume leader: ATLANTIC (108 customers, 49.3% share)
  Conversion leader: E.L.M. LEBLANC (70.6%)

Heat Pump:
  Volume leader: ATLANTIC (155 customers, 44.2% share)
  Conversion leader: DE DIETRICH (50.0%)

AC:
  Volume leader: MITSUBISHI ELECTRIC (197 customers, 66.8% share)
  Conversion leader: E.L.M. LEBLANC (100.0%)

 (main_equipment_category  Boiler  Stove  Other  Heat Pump   AC
  main_brand
  ATLANTIC                    111      0    108        155   76
  MITSUBISHI ELECTRIC           0      0      0        137  197
  MCZ                           0    311      0          0    0
  FRISQUET                    211      0      0          1    0
  E.L.M. LEBLANC              172      0     17          0    1
  DE DIETRICH                 163      0     14          2    0
  SAUNIER DUVAL                89      0      8          0    0
  SOL&MOI                       0      0     68          0    0
  JOLLY MEC                     0     63      0          0    0
  ORANIER                       0     61      0          0    0
  HITACHI                       0      0      4         29   21
  VIESSMANN                    25      0      0         27    0,
  main_equipment_category     Boiler      Stove      Other  Heat Pump  \
  main_brand
  ATLANTIC                 37.837838        NaN  35.185185  42.580645
  MITSUBISHI ELECTRIC            NaN        NaN        NaN  38.686131
  MCZ                            NaN  48.553055        NaN        NaN
  FRISQUET                 45.497630        NaN        NaN   0.000000
  E.L.M. LEBLANC           41.279070        NaN  70.588235        NaN
```

```
DE DIETRICH              43.558282        NaN  50.000000  50.000000
SAUNIER DUVAL            34.831461        NaN  37.500000        NaN
SOL&MOI                        NaN        NaN  17.647059        NaN
JOLLY MEC                      NaN  30.158730        NaN        NaN
ORANIER                        NaN  32.786885        NaN        NaN
HITACHI                        NaN        NaN   0.000000  20.689655
VIESSMANN                48.000000        NaN        NaN  29.629630

main_equipment_category          AC
main_brand
ATLANTIC                  32.894737
MITSUBISHI ELECTRIC       34.517766
MCZ                             NaN
FRISQUET                        NaN
E.L.M. LEBLANC           100.000000
DE DIETRICH                     NaN
SAUNIER DUVAL                   NaN
SOL&MOI                         NaN
JOLLY MEC                       NaN
ORANIER                         NaN
HITACHI                   23.809524
VIESSMANN                       NaN  )
```

```python
import ipynbname
import subprocess
import os
import sys

def export_notebook_to_pdf(overwrite=True, verbose=True):
    """
    Export current notebook to PDF with overwrite control

    Args:
        overwrite (bool): If True, overwrite existing PDF file
        verbose (bool): If True, print progress messages

    Returns:
        bool: True if successful, False otherwise
    """
    try:
        # Get notebook path
        notebook_path = ipynbname.path()
        notebook_path_str = str(notebook_path)
        pdf_path = notebook_path_str.replace('.ipynb', '.pdf')

        if verbose:
            print(f"  Notebook: {notebook_path.name}")
```

```
            print(f"  PDF path: {pdf_path}")

        # Check if PDF exists
        if os.path.exists(pdf_path):
            if overwrite:
                if verbose:
                    print(f"  Removing existing PDF: {pdf_path}")
                os.remove(pdf_path)
            else:
                if verbose:
                    print(f"  PDF already exists and overwrite=False")
                return False

        # Convert to PDF
        if verbose:
            print("  Converting to PDF...")

        result = subprocess.run([
            'jupyter', 'nbconvert', '--to', 'pdf',
            '--no-prompt',  # Suppress input prompts
            notebook_path_str
        ], capture_output=True, text=True)

        if result.returncode == 0:
            if verbose:
                print(f"  PDF saved: {pdf_path}")
            return True
        else:
            if verbose:
                print(f"  Conversion failed:")
                print(result.stderr)
            return False

    except Exception as e:
        if verbose:
            print(f"  Error: {e}")
        return False

# Use the function
if __name__ == "__main__":
    # Overwrite existing PDF
    export_notebook_to_pdf(overwrite=True)

    # Or create new version without overwriting
    # export_notebook_to_pdf(overwrite=False)export_full_notebook_to_pdf()
```

```
Error: Can't identify the notebook path.
```

```python
# Just hardcode your notebook name
notebook_name = "your_notebook_name.ipynb"  # Change this!

import subprocess
import os

pdf_path = notebook_name.replace('.ipynb', '.pdf')
if os.path.exists(pdf_path):
    os.remove(pdf_path)

subprocess.run(['jupyter', 'nbconvert', '--to', 'pdf', '--no-prompt',␣
↪"_eda_price_2025_Q1.ipynb"])
```

```
[NbConvertApp] Converting notebook _eda_price_2025_Q1.ipynb to pdf
[NbConvertApp] Support files will be in _eda_price_2025_Q1_files/
[NbConvertApp] Making directory ./_eda_price_2025_Q1_files
[NbConvertApp] Writing 90256 bytes to notebook.tex
[NbConvertApp] Building PDF
[NbConvertApp] Running xelatex 3 times: ['xelatex', 'notebook.tex', '-quiet']
[NbConvertApp] Running bibtex 1 time: ['bibtex', 'notebook']
[NbConvertApp] WARNING | bibtex had problems, most likely because there were no
citations
[NbConvertApp] PDF successfully created
[NbConvertApp] Writing 2351067 bytes to _eda_price_2025_Q1.pdf
```

```
CompletedProcess(args=['jupyter', 'nbconvert', '--to', 'pdf', '--no-prompt',
'_eda_price_2025_Q1.ipynb'], returncode=0)
```